Examiner only

10. Iodomethane, CH_3I, reacts with a range of nucleophiles. The rate of its reaction with thiosulfate ions, $S_2O_3^{2-}$, has been studied at a range of temperatures.

(a) The reaction is found to be first order with respect to iodomethane and has a rate constant of $3.10\,mol^{-1}\,dm^3\,s^{-1}$ at a particular temperature.

(i) Write the rate equation for this reaction. [2]

....

(ii) The activation energy for this reaction is $81.1\,kJ\,mol^{-1}$ with a frequency factor, *A*, of $5.62 \times 10^{12}\,mol^{-1}\,dm^3\,s^{-1}$. Find the temperature used for this reaction. [3]

T = ... K

Examiner only

(b) One method of performing this experiment involves sampling and quenching.

(i) Explain what is meant by the term 'quenching' and why it needs to be carried out. [2]

(ii) One method of measuring the concentration of thiosulfate ions in solution is by titration using a solution containing iodine. Suggest an indicator for this titration. [1]

(iii) A student suggests using gravimetric analysis to measure the amount of iodide ions produced in the reaction by addition of $Pb^{2+}(aq)$ ions to form a precipitate. Give the colour of this precipitate. [1]

(c) Iodomethane is an intermediate in the conversion of methanol to ethanoic acid.

(i) One catalyst used for this reaction is the iridium-containing complex ion $[Ir(CO)_2I_2]^-$. It is an example of a homogeneous catalyst.

I. State what is meant by a *homogeneous* catalyst. [1]

II. Explain why transition metal complexes such as $[Ir(CO)_2I_2]^-$ can act as homogeneous catalysts. [2]

1410U301 09

Examiner only

(ii) Ethanoic acid is a weak acid with $K_a = 1.76 \times 10^{-5}$ mol dm^{-3}. Calculate the pH of an aqueous solution of ethanoic acid with a concentration of 0.220 mol dm^{-3}. [3]

pH = ...

15